# JULKAR NAIN

**Software Engineer | Full-Stack Developer (Java, Spring Boot, Angular, Next.js) | AI & RAG**

Dhaka, Bangladesh | +880 1300 789158 | julkar929@gmail.com
linkedin.com/in/julkarnain95 | github.com/julkar929

## PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 2+ years of professional experience building secure, high-traffic enterprise web applications in Java (Spring Boot) and Angular for Bangladesh government and public-sector clients, including the Ministry of Education, University of Rajshahi, Gazipur City Corporation, Payra Port and Biman Bangladesh Airlines. Now also contributing to the company AI & Automation R&D team, fine-tuning AI models to extract insights from scanned documents and implementing a Retrieval-Augmented Generation (RAG) system, while expanding into Next.js and NestJS. Strong in RESTful API design, Spring Security (JWT, RBAC), PostgreSQL/Oracle optimization and Docker-based deployment. Former engineering instructor (4+ years) with a B.Sc. in EEE and an M.Sc. in CSE; experienced in code reviews, mentoring and Agile/Scrum delivery.

## TECHNICAL SKILLS

**Languages:** Java, SQL, TypeScript, JavaScript, Python, C, C++, Dart

**Backend:** Spring Boot, Spring Security (JWT, OAuth2, RBAC), Spring Data JPA, Hibernate, RESTful APIs, Microservices, OpenAPI/Swagger, Django, Node.js, Express.js, NestJS

**Frontend:** Angular, HTML5, CSS3, Bootstrap, Thymeleaf, Tailwind CSS, Material Design, React, Next.js

**AI & Automation:** Generative AI, AI model fine-tuning, Retrieval-Augmented Generation (RAG), data extraction from scanned documents, Python

**Databases:** PostgreSQL, Oracle, MySQL, MongoDB, SQLite, Redis; schema design, indexing and query optimization

**DevOps & Tools:** Docker, Kubernetes, Git (GitHub, GitLab, Bitbucket), CI/CD (GitLab CI, GitHub Actions), Maven, Postman, Jira, SLF4J

**Testing & Reporting:** JUnit, Mockito, Jasmine; JasperReports, JasperSoft Studio

**Practices:** Agile/Scrum, code review, secure application development, API integration, debugging, AI-assisted development

## PROFESSIONAL EXPERIENCE

**Software Engineer** Oct 2024 – Present
*Addie Soft Limited, Dhaka, Bangladesh*

- Design and develop secure RESTful APIs and full-stack enterprise modules using Java, Spring Boot, Spring Data JPA, Hibernate and Angular for high-concurrency national platforms.
- Member of the AI & Automation R&D team: fine-tune AI models to extract useful insights from scanned documents efficiently, and implement a Retrieval-Augmented Generation (RAG) system to make automation workflows smarter and more reliable in upcoming company software.
- Develop a new core company project using modern web technologies (Next.js, NestJS).
- Implement authentication and authorization with Spring Security, JWT/OAuth2 and role-based access control (RBAC) across multiple modules.
- Build responsive, reusable UI components with Angular, HTML, CSS and Bootstrap, integrated with backend REST APIs.
- Design and optimize PostgreSQL schemas, complex SQL queries and indexes; refactor legacy queries to remove timeouts during peak-traffic periods.
- Generate dynamic PDF/Excel reports with JasperReports and Thymeleaf and support bulk data-processing workflows for management decision-making.
- Deploy applications with Docker, support GitLab CI/CD pipelines and implement centralized logging with SLF4J.
- Write unit and integration tests (JUnit, Mockito), conduct code reviews, and mentor 3 junior developers.

**Remote Instructor** Dec 2022 – Dec 2024
*Possible Edu Care (Part-time, Remote)*

- Delivered online technical training on full-stack development with Java, Spring Boot and Angular.

**Instructor (Electrical & ICT)** Aug 2018 – Jan 2023
*Rajshahi Institute of Technology, Rajshahi, Bangladesh*

- Taught Computer Programming (Java, Python), Microcontrollers, PLC and Electrical Engineering to diploma students; supervised hands-on embedded-systems projects.

## PROFESSIONAL PROJECTS (ADDIE SOFT LIMITED)

**EMIS Core Web Applications & Central Portal Hub** 2024 – 2026
*Directorate of Secondary and Higher Education (DSHE), Ministry of Education | apps.emis.gov.bd/portal*

- National platform with Single Sign-On (SSO), administrative dashboards, automated teacher transfers and payroll profiles for thousands of institutions.
- Built end-to-end features in Angular and Spring Boot (Oracle, PostgreSQL), including PDS update and iBAS integration, and implemented transactional logic for automated BCS cadre transfers and MPO EFT updates under high concurrency.

**Rajshahi University Examination Result Portal** 2025 – 2026
*University of Rajshahi | eresult.ru.ac.bd*

- Academic result processing and publication portal serving thousands of students during semester and annual result announcements.
- Developed core full-stack features, refactored SQL queries and indexes to eliminate peak-hour timeouts, and resolved complex API integration bugs to improve stability and uptime.

**GCC Water Supply Billing & New Connection Management System** 2025 – 2026
*Gazipur City Corporation (GCC) | water.gcc.gov.bd*

- Citizen-facing billing application for new water connection requests, ledger management, digital bill generation and online payment.
- Implemented feature enhancements from municipal requirements and user feedback; improved responsive layouts for mobile and tablet.

**Other client work:** Custom software for Payra Port and Biman Bangladesh Airlines (Addie Soft Limited).

## PERSONAL PROJECTS

**BookNook – Online Library** | Next.js, Express.js, MongoDB, JWT, BetterAuth, Tailwind CSS
*Live: booknook-abc.vercel.app | Code: github.com/julkar929/booknook-clients, github.com/julkar929/booknook-servers*

- Built a full-stack library app (browse by genre, borrow/return, curator CRUD dashboard) with a separate Express.js REST API; atomic MongoDB $inc stock updates and a no-double-borrow rule.
- Implemented JWT and Google login with protected routes and a dark/light theme.

**BazarDor (Grocery Price Tracker)** | Next.js 15, TypeScript, MongoDB, BetterAuth, DaisyUI
*Live: bazardor-orpin.vercel.app | Code: github.com/julkar929/bazardor*

- Bangla-first app with live price ticker, daily risers/fallers, sortable category pages and login-protected product details with division-wise prices (email, Google and GitHub login).

**FitLog (Workout Library)** | Next.js, TypeScript, Tailwind CSS
*Live: myfitlogs.vercel.app | Code: github.com/julkar929/fitLog*

- Responsive gym companion that fetches 12 lifts from an API, builds a daily plan and saved list, and tracks minutes and calories (persisted in localStorage).

**DristiScan (Experimental R&D)** | Gemini, Gemma 3 27B, OCR, RAG, PostgreSQL, REST API
*Live: julkarocr.vercel.app*

- AI marksheet scanner that extracts handwritten roll numbers and marks, stores results in PostgreSQL, exposes a Swagger-documented REST API, and includes a RAG teacher assistant with a Bangla/English UI.

**Earlier projects:** Student Management and E-Commerce (Spring Boot, Angular, PostgreSQL); E-Medicine Shop (Django); Bazar Accountant (Flutter); ExamTime Controller (AVR); Home Automation (Arduino, IoT).

## EDUCATION

**M.Sc. in Computer Science & Engineering** 2021 – 2023
*University of Rajshahi | CGPA: 3.08 / 4.00*

**B.Sc. in Electrical & Electronic Engineering** 2013 – 2017
*Varendra University | CGPA: 3.78 / 4.00*

## CERTIFICATIONS & TRAINING

- Generative AI Mastermind – Outskill (Certificate of Completion)
- Web Development (Next.js, NestJS, Express.js, MongoDB) – Programming Hero, Batch 14
- Java Certification Course – Programming Hub (2024)
- 21st Century Employability Skills Program (Advanced) – Wadhwani Foundation (2023)
- Docker Certification Course – Programming Hub (2022)
- Professional Web & Content Management – Bangladesh Hi-Tech Park Authority / Raj IT Solution Ltd. (2017)

## LANGUAGES

Bengali (Native), English (Professional)